

Quadratic equations: what we can already solve

ANSWER KEY • FOR TEACHERS



Interactive version

REMEMBER

From grade 8

Level 1 • threshold g8.inc-quad

1.

Solve $2x^2 = 50$. Give both roots, separated by a semicolon.**Answer: 5; -5**

Level 1 • threshold g8.inc-quad

2.

Solve $x^2 - 6x = 0$.**Answer: 0; 6**

Level 1 • threshold g8.inc-quad

3.

Which of these numbers is a root of $x^2 - x - 12 = 0$?**A** 3**B** 12**C** -4

D $-3 - (-3)^2 -$
 $(-3) - 12 = 9 +$
 $3 - 12 = 0.$

WORKED EXAMPLES

Worked examples

YOUR TURN

Level 1 • threshold g9.qe.1

Write $4 - x^2 = 3x$ in the form $ax^2 + bx + c = 0$ with $a > 0$. Enter a , b and c , in this order, separated by semicolons.

WORKED SOLUTION

 $4 - x^2 - 3x = 0$, so $-x^2 - 3x + 4 = 0$. Multiplying by -1 gives $x^2 + 3x - 4 = 0$: $a = 1$, $b = 3$, $c = -4$.**Answer:** $a = 1$, $b = 3$, $c = -4$

YOUR TURN

Level 1 • threshold g9.qe.2

Solve $(3x + 2)(x - 5) = 0$.**Answer:** $x_1 = -\frac{2}{3}$, $x_2 = 5$

YOUR TURN

Level 1 · threshold

g9.qe.2

Solve $(x - 3)^2 = 25$.

Answer: 8; -2

CHECK

Check yourself

Level 1 · threshold

g9.qe.1

1.

Which of these are quadratic equations? Choose all that apply.

☐ A $\frac{4}{x^2} - 1 = 0$

☐ B $3^2x + x - 4 = 0$

☒ C $x^2 - 7x = 0$ — It is quadratic ($a = 1$, $b = -7$) and incomplete, because $c = 0$.

☐ D $5 - 2x^2 = 0$ — It is quadratic: $-2x^2 + 5 = 0$, with $a = -2$, $b = 0$, $c = 5$. The order of the terms does not matter.

Level 2 · satisfactory

g9.qe.1

2.

Which of these is an **incomplete** quadratic equation?

☐ A $x^2 + x = 1$

☐ B $(x - 1)^2 = 0$

☒ C $3x^2 = 5x - 3x^2 - 5x = 0$: $c = 0$.

☐ D $2x^2 + 3x - 5 = 0$

Level 1 · threshold

g9.qe.1

3.

Write $7 - 3x^2 = 2x$ in the form $ax^2 + bx + c = 0$ with $a > 0$. Enter a , b and c , in this order.

WORKED SOLUTION

$7 - 3x^2 - 2x = 0$, so $-3x^2 - 2x + 7 = 0$. Multiplying by -1 gives $3x^2 + 2x - 7 = 0$: $a = 3$, $b = 2$, $c = -7$.

Answer: $a = 3$, $b = 2$, $c = -7$

Level 1 · threshold

g9.qe.2

4.

Solve $(x + 6)(3x - 1) = 0$.

Answer: $x_1 = -6$, $x_2 = \frac{1}{3}$

Level 1 · threshold

g9.qe.2

5.

Solve $(x + 4)^2 = 36$.

Answer: 2; -10

6.

Solve $(x - 5)^2 = -9$.

- ☐ A $x = -4$
☒ B **no real roots** — A square is never negative, so no real number works.
 ☐ C $x = 5$
☐ D $x = 8$ or $x = 2$

7.

Ona solves $x(x - 3) = 4$ like this: " $x = 4$ or $x - 3 = 4$, so $x = 4$ or $x = 7$." What is wrong?

- ☒ A **The zero-product rule needs 0 on one side. Here it is 4, so the rule does not apply, and $x = 7$ is not a root.** — Check: $7 \cdot (7 - 3) = 28 \neq 4$. And her reasoning does not show that 4 is a root either: $4 \cdot 1 = 4$ happens to work, but not because of the rule.
 ☐ B She should divide both sides by x .
 ☐ C Nothing: 4 and 7 are both roots.
 ☐ D She should have written $x = 0$ or $x - 3 = 0$.

EXAM TASK

A hidden square

Level 3 · basic

PUPP style · 3 points

g9.qe.2

The equation $x^2 - 2x + 1 = k$ is given, where k is a number.

- Write the left side as the square of a binomial.
- Solve the equation when $k = 49$.
- For which values of k does the equation have no real roots? Explain.

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

- | | |
|---|------|
| a) $x^2 - 2x + 1 = (x - 1)^2$. | 1 pt |
| b) $x - 1 = 7$ or $x - 1 = -7$, so $x = 8$ or $x = -6$. | 1 pt |
| c) $k < 0$, because $(x - 1)^2 \geq 0$ for every x . | 1 pt |

WORKED SOLUTION

- a) $x^2 - 2x + 1 = x^2 - 2 \cdot x \cdot 1 + 1^2 = (x - 1)^2$.
- b) $(x - 1)^2 = 49$, so $x - 1 = 7$ or $x - 1 = -7$: $x = 8$ or $x = -6$. Check: $8^2 - 2 \cdot 8 + 1 = 49$ ✓ and $(-6)^2 - 2 \cdot (-6) + 1 = 49$ ✓.
- c) The equation is $(x - 1)^2 = k$. A square is never negative, so for $k < 0$ there are no real roots. (For $k = 0$ there is one root, $x = 1$; for $k > 0$ there are two.)

Answer: a) $(x - 1)^2$; b) $x = 8$ or $x = -6$; c) $k < 0$.

TEACHER NOTES

For the teacher

Pupils can say: “I can write a quadratic equation as $ax^2 + bx + c = 0$ and read a, b, c with their signs; I can tell a complete equation from an incomplete one; I can solve a product equal to zero and a square equal to a number.”

Where this lesson sits. Since V-594, the definition of a quadratic equation and the incomplete equations are grade 8 content (31.2.2). This lesson collects that knowledge, names the general form, and adds two easy types: a product of linear factors equal to 0, and $(x + p)^2 = d$. The hook sets up lesson 1.2: (E) became easy because it could be rewritten as (D). Can that always be done?

Common mistakes and what helps

- Reading b and c before moving the terms to one side ($7 - 3x^2 = 2x$ read as $b = 2$). Insist on “... = 0” first.
- Signs in the zero-product rule: $x + 6 = 0$ gives $x = -6$; $3x - 1 = 0$ gives $x = \frac{1}{3}$, not 3.
- Using the rule when the right side is not 0 (item c7). Ask: “Why does the rule need a 0?”
- Forgetting the negative case in $(x + p)^2 = d$ (c5), or “taking the root” of a negative number (c6).
- Dividing by x , which loses the root $x = 0$ (r2). Lesson 1.7 comes back to this.
- Calling an equation incomplete just because it is short ($x^2 + x = 1$), or because it is easy to solve ($(x - 1)^2 = 0$).

Key terms (EN • LT): quadratic equation • *kvadratinė lygtis*; coefficient • *koeficientas*; constant term • *laisvasis narys*; complete / incomplete equation • *pilnoji / nepilnoji lygtis*; root • *sprendinys*; zero-product rule • *sandauga lygi nuliui, kai bent vienas dauginamasis lygus nuliui*.

Timing (45 min):

- Remember: 5 min.
- Hook: 3 min, one try.
- Explain: 12 min.
- Examples and “Your turn”: 13 min.
- Check c1–c6: 10 min.

c7 (level 3) and the exam task are for pupils who finish early, or for homework. The exam task links this lesson to the next ones: its left side can already be recognised as a square (lesson 1.2 makes one), and the sign of k decides the number of roots (lesson 1.5 does the same with D).

Programme: 32.1.1.1 “*Kvadratinės lygtys*”, bridging from 31.2.2 “*Apibrėžiamos sąvokos: antrojo laipsnio (kvadratinė) lygtis su vienu nežinomuoju ir jos sprendinys.*” NŠA threshold minimum (2025): 2.2.1 (incomplete equations) and 2.3.2 (“Which equations are quadratic?”, like c1). Achievements A1, B2.